

THE SOUTH AFRICAN ENERGY SECTOR REPORT 2023



**DIRECTORATE: ENERGY ECONOMICS
AND STATISTICS**



**mineral resources
& energy**

Department:
Mineral Resources and Energy
REPUBLIC OF SOUTH AFRICA



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DIRECTORATE: ENERGY ECONOMICS AND STATISTICS

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FOREWORD

It is with the greatest appreciation and honour to introduce the 2023 South African Energy Sector Report. This report presents us with an opportunity to inform and update all the relevant stakeholders about the latest energy demand and supply trends in South Africa as prescribed in the National Energy Act no.8 of 2008. The report is based on the 2021 Balance compiled by the Department of Mineral Resources & Energy (DMRE) and mainly focusses on analysing energy supply and the subsequent consumption broken down into various sectors of the economy.

This report is published annually with data and analysis based on the latest published South African Energy Balances. We have made remarkable progress in improving the overall quality of our energy data to ensure that accurate, timely and reliable data is provided in all our publications and hope that this report will become a source of reference among energy analysts in South Africa and abroad. We will continuously collaborate with all our data providers to ensure that our energy data is of highest quality possible.

I would like to extend my most sincere thanks and appreciation to the Energy Economics and Statistics Directorate for the hard work that went into the compilation of this publication. I would also like to extend my appreciation to all the energy data providers and stakeholders for their concerted efforts and support during the process of compiling this report. Comments and inputs are welcome and could be addressed to Publications@energy.gov.za.

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Mr. Jacob Mbele

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INTRODUCTION

The South African energy sector stands at the crossroads of transformative change, aimed at addressing both historical challenges and future opportunities. As the largest economy on the African continent, South Africa's energy landscape is crucial not only for its own development but also for shaping the broader regional and continental energy dynamics. The sector is undergoing a profound shift driven by a combination of factors, including the imperative to diversify the energy mix, enhance energy security and mitigate environmental impacts.

Historically, South Africa has been heavily reliant on coal for its energy needs, a legacy of its abundant coal reserves. The country has taken Policy initiatives aimed at incorporating renewable energy sources, such as wind, solar and hydroelectric power, into its energy mix. The transition aligns with global trends emphasizing clean and sustainable energy solutions to address climate change concerns.

The integrated resource plan envisages a diversity of energy sources, including coal, gas and storage. The proportion of coal in the energy mix will decline, and the proportion of renewable energy will increase. According to the National Development Plan, it is envisaged that by 2030, South Africa will have an environmentally sustainable society, climate change resilience, an expanded low-carbon economy, and reduced emissions.

In this report, the 2021 South African Energy Balance is analyzed by breaking it down per sub-sectors, that is, the total primary energy supply, which includes domestic production as well as imports and exports and total final consumption which includes the economic sectors, industry, transport, agriculture, commercial and public as well as residential sectors. There is also a further breakdown into the various industry subsectors in terms of energy consumption. The purpose of this report is therefore to show how and in what proportions the country's available energy is supplied and consumed by different sectors.

LIST OF ABBREVIATIONS

bbl/d	Barrels per day
CAIA	Chemical and Allied Industries' Association
CEF	Central Energy Fund
CTL	Coal-To-Liquid
DMRE	Department of Mineral Resources & Energy
DMRE	Department of Energy
GDP	Gross Domestic Product
GTL	Gas-To-Liquid
IEP	Integrated Energy Plan
INEP	Integrated National Electrification Programme
IRP	Integrated Resource Plan
JPoI	Johannesburg Plan of Implementation
LPG	Liquified Petroleum Gas
Mt	Megaton
MW	Megawatts
MYPD	Multi-Year Pricing Determination
NERSA	National Energy Regulator of South Africa
PetroSA	Petroleum Oil and Gas Corporation of South Africa
TJ	Tera Joules
R/t	Rand per Ton
REIPPPP	Renewable Energy Independent Power Producer Procurement Programme
ROM	Run-of-Mine
SAPIA	South African Petroleum Industry Association
StatsSA	Statistics South Africa
SWH	Solar Water Heaters
WSSD	World Summit on Sustainable Development



KEY ENERGY POLICIES AND LEGISLATIONS

1. White Paper on the Energy Policy, December 1998

The White Paper on the Energy Policy was developed so as to clarify government policy regarding the supply and consumption of energy for the next decade. It was intended to address all elements of the energy sector as practically as it could. This White Paper gives an overview of the South African energy sector's contribution to GDP, employment, taxes and the balance of payments. It concludes that the sector can greatly contribute to a successful and sustainable national growth and development strategy.

The main objectives of the White Paper are the following:-

- Increasing access to affordable energy services.
- Improving energy governance.
- Stimulating economic development.
- Managing energy-related environmental impacts.
- Securing supply through diversity.

2. White Paper on Renewable Energy, November 2003

The White Paper on Renewable Energy supplements the Government's overarching policy on energy as set out in its White Paper on the Energy Policy (as stated above), which pledges 'Government support for the development, demonstration and implementation of renewable energy sources for both small and large-scale applications'.

This White Paper sets out Government's vision, policy principles, strategic goals and objectives for promoting and implementing renewable energy in South Africa. Additionally, it has the following two goals:

- to inform the public and the international community of the Government's goals, and how the Government intends to achieve them, and;
- to inform Government agencies and Organs of State of these goals, and their roles in achieving them.

3. Nuclear Energy Policy, October 2008

The nuclear sector in South Africa is mainly governed by the Nuclear Energy Act 1999, Act 46 of 1999 and National Radioactive Waste Disposal Institute Act, Act 53 of 2008. National Nuclear Regulator (NNR) Act 1999, Act 47 of 1999. The Cabinet approved the Nuclear Energy Policy for South Africa in October 2008. The Nuclear Energy Policy outlines the South African government's vision for the development of an extensive nuclear energy programme by ensuring that Government's objective on the prospecting and mining of uranium ore and the use of uranium (or other relevant nuclear materials) as a primary resource of energy must be regulated and managed in a manner that will be for peaceful purposes.

Through the Nuclear Energy Policy, Government aims to achieve the following objectives:

- Promotion of nuclear energy as an important electricity supply option through the establishment of a national industrial capability for the design, manufacture and construction of nuclear energy systems;
- Establishment of the necessary governance structures for an extended nuclear energy programme;
- Creation of a framework for safe and secure utilization of nuclear energy with minimal environmental impact;
- Contribution to the country's national programme of social and economic transformation, growth and development;
- To guide in the actions to develop, promote, support, enhance, sustain and monitor the nuclear energy sector in South Africa;
- Attainment of global leadership and self-sufficiency in the nuclear energy sector in the long-term;
- Exercise control over unprocessed uranium ore for export purposes for the benefit of the South African economy;
- Establishing of mechanisms to ensure the availability of land (nuclear sites) for future nuclear power generation;
- Allow for the participation of public entities in the uranium value chain;
- Promoting energy security for South Africa;
- Improvement of the quality of human life and to support the advancement of science and technology;

- Reduction of greenhouse gas emissions; and
- Skills development related to nuclear energy.

4. Integrated Resource Plan (IRP) 2010-30

The IRP is a generation capacity expansion plan based on a least-cost electricity supply and demand balance in the long term and incorporates government policy. The IRP review presents an analysis of broad options at the disposal of the country with the ultimate view to meet three distinct but not mutually exclusive aspirations, namely, security of supply, energy affordability and carbon emissions reduction. There has been several developments in the electricity industry in South Africa since the promulgation of IRP 2019 and these in the main include:

- The establishment of the Presidential Climate Commission (PCC) which is a multi-stakeholder body established by the President of the Republic of South Africa to advise on the country's climate change response and pathways to a low-carbon climate-resilient economy and society. One of its main outputs, was the development of the Just Transition Framework. In addition, the Inter-Ministerial Committee (IMC) for Just Energy Transition developed the Just Energy Transition Investment Plan (JET IP) and the subsequent Implementation Roadmap for the period 2023-2027 which sets out the scale of need and the investments required to achieve the decarbonisation commitments in line with South Africa's Nationally Determined Contribution (NDC).
- The Energy Action Plan is designed to reduce load-shedding and achieve energy security as announced by the President on 25, July 2022. The plan implemented under the supervision of the National Energy Crisis Committee (NECOM) consists of five crucial actions including; improving the availability of existing power generation plants, enabling and accelerating private investment in power generation capacity, accelerating procurement of new capacity from renewables, gas and battery storage, unleashing businesses and households to invest in rooftop solar, and fundamental transformation of the electricity sector to achieve long-term energy security.
- The removal of licensing requirements for the development of power generation by consumers (embedded generation) meant to enable and attract more investment necessary to address load shedding. This intervention has to date resulted in the registration of projects with a potential generation capacity of over 6000 MW.

- The intervention by the National Treasury to address the Eskom debt which impacts on the utility's ability to sustain operations and invest in additional infrastructure. Addressing Eskom debt effectively will enable much needed investment in critical transmission and other infrastructure, and proper maintenance of plant and equipment.
- Procurement of approximately 6000 MW of additional generation capacity under the technology agnostic Risk Mitigation Independent Power Producer Procurement Programme (RMIPPPP), Bid Window 5 and 6 of Renewable Energy and Bid Window 1 of Battery Storage.

The Integrated Resource Plan (IRP) is a living plan that is expected to be continuously revised and updated as necessitated by changing circumstances. It was first promulgated in March 2011, and the last review of the IRP was done in 2019.

The Department of Mineral Resources & Energy has now developed a review of the IRP. The Minister of Mineral Resources & Energy, Mr. Gwede Mantashe has under section 4 (1) of the Electricity Regulations on New Generation Capacity published the Integrated Resource Plan, 2023 for public comments in the government gazette and Interested persons and organisations were invited to submit their comments by 23 February 2024.

5. National Energy Act, 2008

The National Energy Act, 2008 (Act 34 of 2008) ensures that diverse energy resources are available in sustainable quantities and at affordable prices in South Africa. In addition, the Act provides for the increased use of renewable energies, contingency energy supplies, the holding of strategic energy feedstock and carriers, and adequate investment in energy infrastructure.

6. Petroleum Products Act, 1977

The aim of the Petroleum Products Act, 120 of 1977, is to:

- Provide measures in the saving of petroleum products and an economy in the cost of the distribution thereof, the maintenance and control of a price, for the furnishing of certain information regarding petroleum products, and for the rendering of services of a particular standard, in connection with petroleum products;

- Provide for the licensing of persons involved in the manufacturing and sale of certain petroleum products;
- Promote the transformation of the South African petroleum and liquid fuels industry;
- Provide for the promulgation of regulations relating to such licenses; and
- Provide for matters incidental.

7. Nuclear Energy Act, 1999

The aims of the Nuclear Energy Act, of 1999 (Act 46 of 1999) are:

- To provide for the establishment of the National Energy Corporation of South Africa (Necsa) and define its functions, powers, financial and operational accountability, governance and management;
- To provide for responsibilities for the implementation and application of the Safeguards Agreement and any additional protocols entered into by the Republic of South Africa and the International Atomic Energy Agency in support of the Nuclear Non-Proliferation Treaty acceded to by the Republic;
- To regulate the acquisition and possession of nuclear fuel, certain nuclear and related material and certain related equipment, as well as the importation and exportation of, and certain other acts and activities relating to, that fuel, material and equipment in order to comply with the international obligations of the Republic;
- To prescribe measures regarding the discarding of radioactive waste and the storage of irradiated nuclear fuel, and to provide for incidental matters.

8. The Gas Act, 2001

The aims of the Gas Act are as follows:

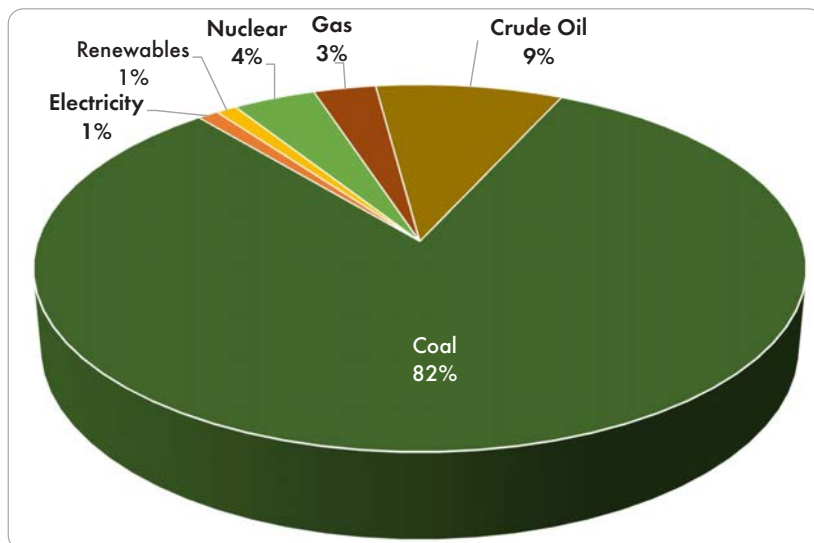
- To promote the orderly development of the piped gas industry;
- To establish a national regulatory framework;
- To establish a National Gas Regulator as the custodian and enforcer of the national regulatory framework; and
- To provide for matters connected therewith.



1. ENERGY SUPPLY

The South African energy supply is dominated by coal which made up 82% of the primary energy supply in 2021, followed by crude oil at 9% and Nuclear at 4%. Natural gas contributed 3% while Renewables and Electricity contributed 1% each respectively to the total primary supply during the same period. The primary energy supply in this case includes indigenous production and imported sources less exported quantities. (Figure 1 below).

Figure 1: Total primary energy supply, 2021



Source: DMRE Energy Balances, 2021

1.1 Crude Oil and Petroleum Products

Overview

In 2021, Egypt had the largest oil refinery capacity among the African countries, amounting to 833,000 barrels per calendar day. South Africa downgraded to the fourth largest country with the largest oil refining capacity amounting to 538,000 per day (Statista.com:2022). There are six refineries in the country; four of the refineries are on the coast and two are inland.

There is a Coal-To-Liquid refinery of 150 000 barrels per day and in addition, there is a gas-to-liquid refinery with a capacity to produce 45 000 barrels per day. Three crude oil refineries have not been in operation for the past 2 years. Petro Sa's Mossel Bay refinery closed because of being unable to obtain feedstock to stay in operation. The Engen refinery had a major fire which caused extensive damage. Astron Energy is still committed to the option of restarting its refinery soon (DMRE, 2023). According to DMRE Energy Balances, Crude Oil imports have significantly dropped in recent years due to the Ukraine war causing supply challenges. Therefore, South Africa is becoming dependent on imports of finished Oil products.

Sasol uses both the Coal-To-Liquids (CTL) and Gas-To-Liquids (GTL) technologies. The Petroleum Oil and Gas Corporation of South Africa (PetroSA) produces synthetic products using GTL technology. Major refineries include Sapref and Enref in Durban, Astron Energy in Cape Town, and Natref at Sasolburg. The capacity and location of the six refineries are shown in Table 1-1 below.

Table 1: Refinery production capacity

Refinery	Area and Province	Type	Capacity*
Astron Energy	Cape Town, Western Cape	Crude	100 000
Enref	Durban South, KwaZulu Natal	Crude	0
Natref (Sasol Oil owns 64% and Total SA 36%)	Sasolburg, Free State	Crude	108 000
PetroSA	Mossel Bay, Western Cape	Synthetic (GTL)	0
Sasol Synfuels	Secunda, Mpumalanga	Synthetic (CTL)	150 000
Sapref (BP owns 50% and Shell 50%)	Durban South, KwaZulu Natal	Crude	180 000
Total			538 000

Source: SAPIA Annual Report 2021

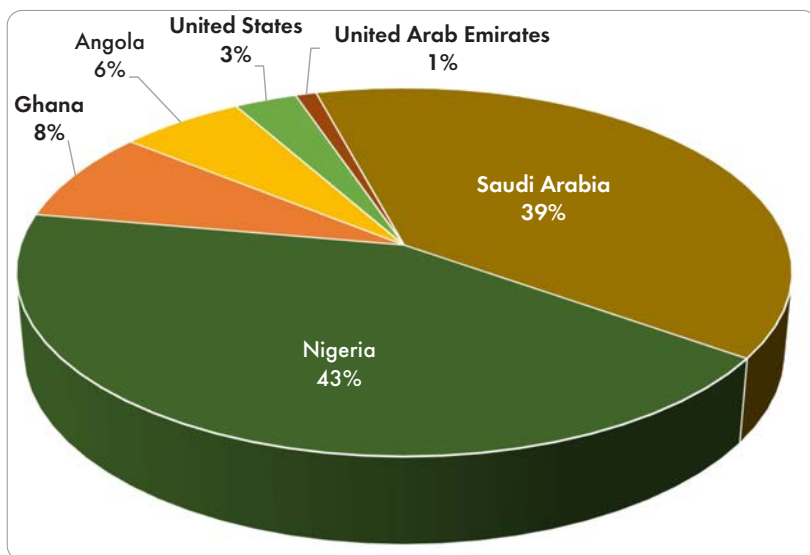
*Crude equivalent (bbl/day)

Crude Oil

South Africa continued to import almost 100% of its crude oil requirements in 2021. The total primary crude oil supply was used by refineries during the transformation process for liquid fuels production. Almost all the imported crude oil is used to produce liquid fuels, with a small percentage used towards lubricants, bitumen, solvents and other petrochemicals.

As depicted in Figure 2 and Table 2 below, crude oil imports were mostly from OPEC countries in 2021, with 43% imported from Nigeria followed by Saudi Arabia (39%), Ghana (8%), Angola (6%), United Arab Emirates and United States at 1% and 3%, respectively and small volumes from various producers (SARS trade data, 2021).

Figure 2: Crude oil imports by country of origin, 2021



Source: South African Revenue Service (SARS), 2021

Table 2: Crude Oil imports volumes by country of origin, 2021

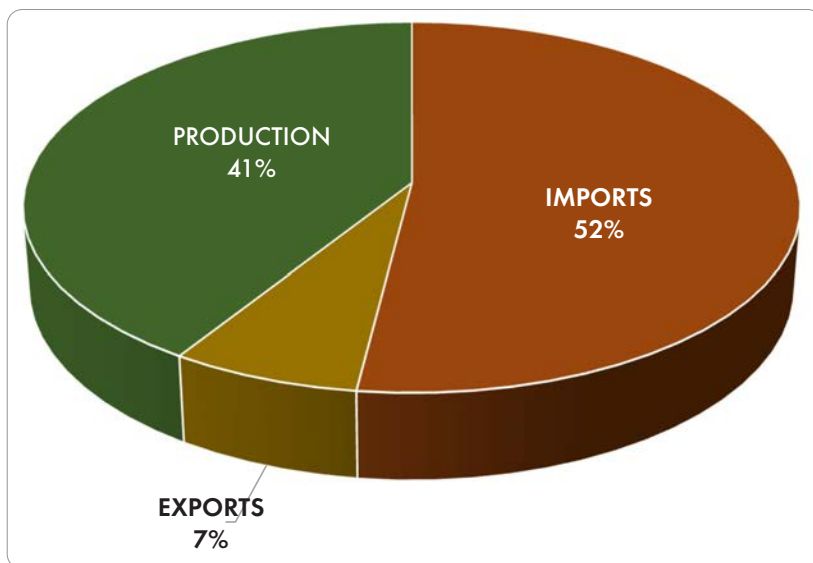
Country of Origin	Imports volume	
	Tons	%
Angola	662 027	5,71
Ghana	947 785	8,17
Nigeria	4 964 460	42,78
Saudi Arabia	4 440 622	38,27
United Arab Emirates	129 961	1,12
United States	365 599	3,15
Others	93 840	0,81
Grand Total	11 604 294	100

Source: South African Revenue Service (SARS), 2021

Petroleum products

According to the 2021 Energy Balances, 41% of the total petroleum products supply was produced locally. Imports amounted to 52% to make up for the local production shortfall while exports amounted to 7% (Fig. 3).

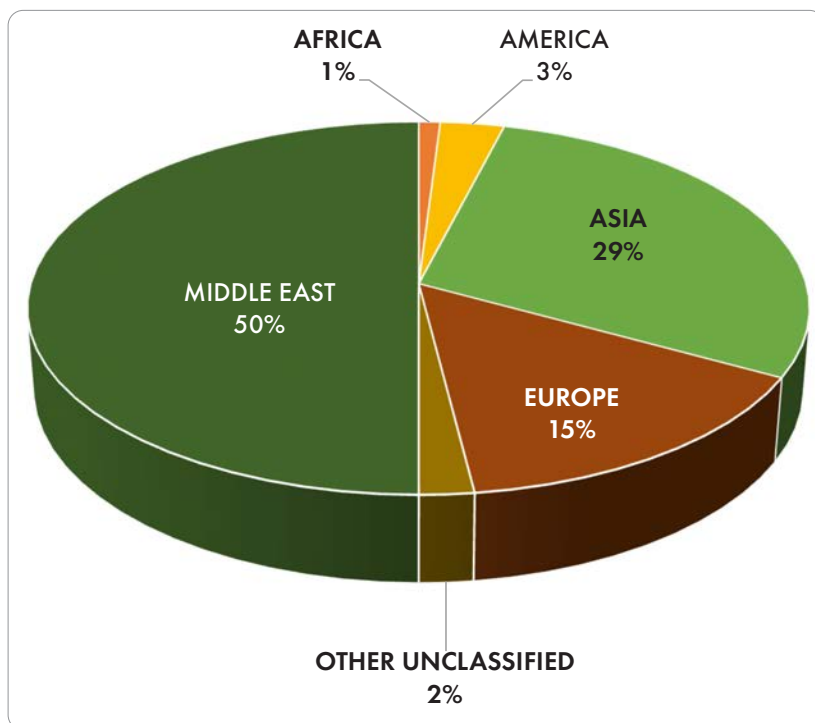
Figure 3: Petroleum products supply, 2021



Source: DMRE Energy Balances, 2021

Most of the petroleum products imported into South Africa were from the Middle East at 50% as depicted in Figure 4 below. The Asian area, which was dominated by India, Malaysia, and Singapore, came in second and third, respectively, at 29%. 15% of the products imported came from Europe, with the remaining regions coming from America (3%), Africa (1%), and other regions.

Figure 4: Petroleum products imports by region, 2021



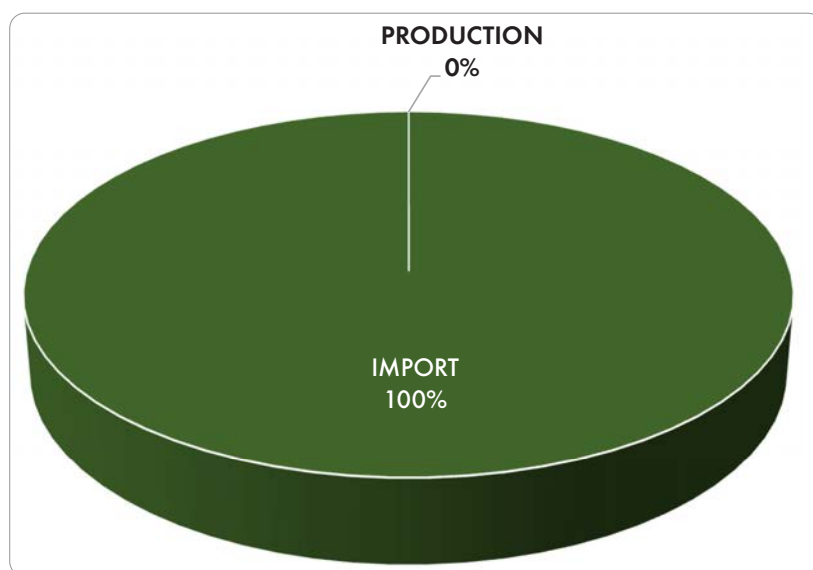
Source: DMRE Energy Balances, 2021

1.2 Natural gas

Currently, natural gas is imported into South Africa by Sasol Gas via an 865 km pipeline from the Temane and Pande gas fields in Mozambique. Reserves in the Temane and Pande are estimated at around 2.6 trillion cubic feet (TCF). The pipeline has a capacity of 240 million gigajoules (GJ) per annum. Approximately 120 million GJ is used annually by Sasol in the GTL and chemicals plant in Secunda, while the balance is distributed to commercial and industrial customers via a pipeline network covering more than 3,000 km in the Mpumalanga, Free State, Gauteng, Kwa-Zulu Natal and Northwest.

In 2021, natural gas made up 3% of the total primary energy supply in South Africa. South Africa imported all of its Natural gas requirements with no exports in 2021. No Natural gas was produced locally as per Figure 5.

Figure 5: Natural gas primary supply, 2021



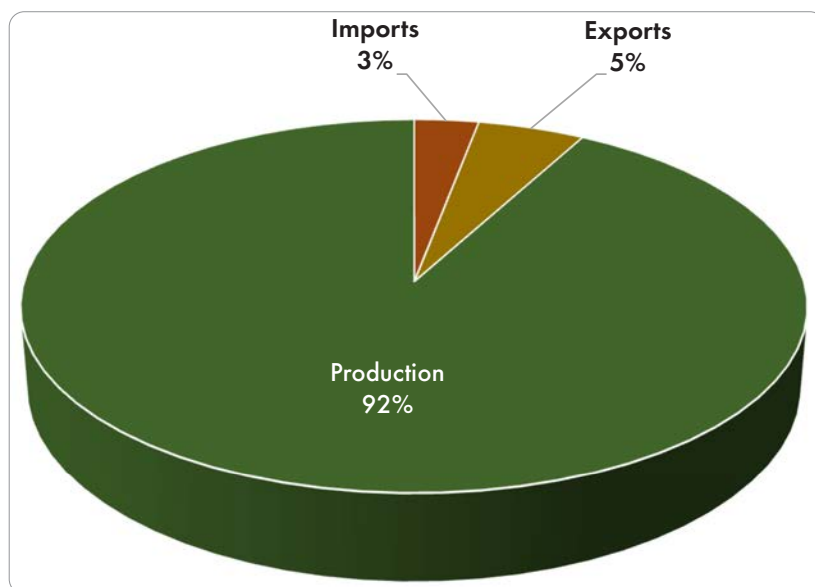
Source: DMRE Energy Balances, 2021

1.3 Electricity

Eskom is a vertically integrated, state-owned power company that owns and operates the National electricity grid and supplies about 90% of the power generation in South Africa. Eskom generates 30% of the electricity that is used in the rest of Africa, respectively exported to neighbouring countries such as Botswana, Eswatini, Lesotho, Mozambique, Namibia, Zambia, and Zimbabwe (Eskom, 2021).

The electricity generated by Eskom and IPPs, combined with imports from neighbouring countries, is supplied in bulk to distributors, i.e., large metros and other municipalities and also distributed to industrial, commercial, residential and others (Eskom, 2021). It owns and operates several coal-fired, gas-fired, hydro and pumped storage power stations, as well as one nuclear power station. Local production amounted to 92% of the total electricity supply in 2021, while the country's imports and exports amounted to 3% and 5%, respectively.

Figure 6: Electricity supply, 2021

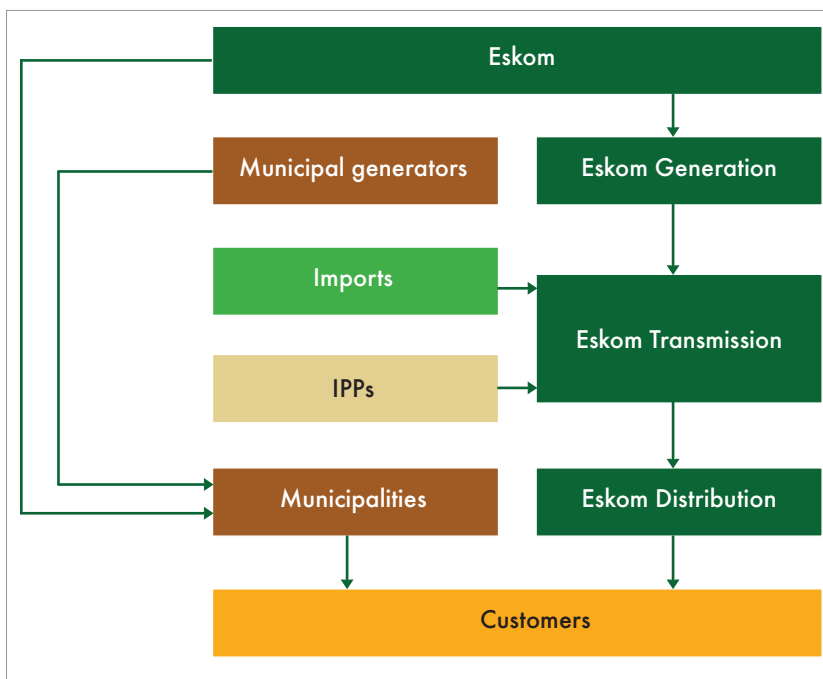


Source: DMRE Energy Balances, 2021

Generation, Transmission and Distribution

Electricity infrastructure comprises three sub-sectors, namely: - generation, transmission and distribution. In terms of generation, Eskom dominates the production of electricity. Eskom generates, transmits and distributes electricity to industrial, mining, commercial, agricultural and residential customers in South Africa, and to municipalities, who in turn redistribute electricity to businesses and households within their areas. The utility also purchases electricity from Independent Power Producers (IPP's) in terms of various agreement schemes as well as electricity generating facilities beyond the country's borders. Most power stations are located in Mpumalanga, except for Lethabo and Matimba which are located in the Free State and Limpopo provinces respectively.

Figure 7: Electricity Generation, Transmission and Distribution

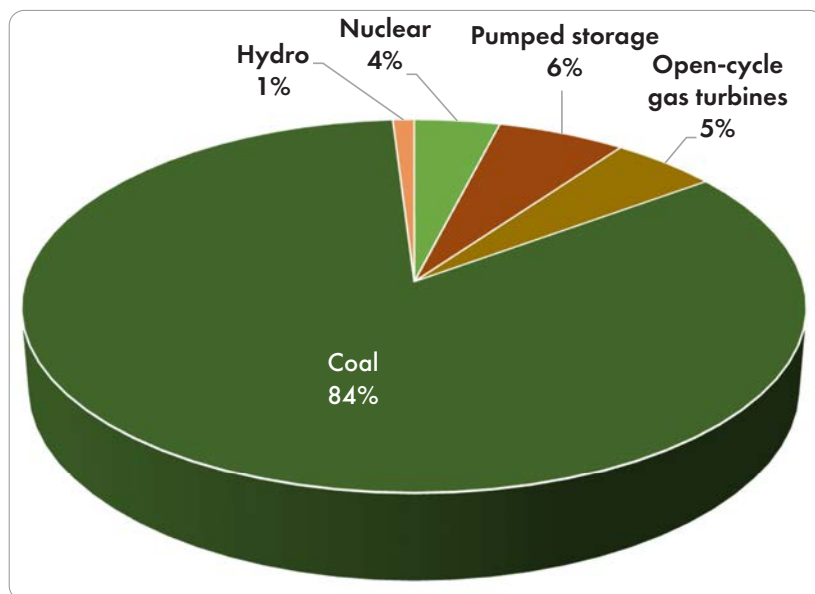


Source: Eskom:2021

Eskom Generating Capacity

According to Eskom's Integrated Results 2021 the total nominal capacity as of March 2021 amounted to 47 GW.

Figure 8: Power Station Maximum Generating Capacity Mix



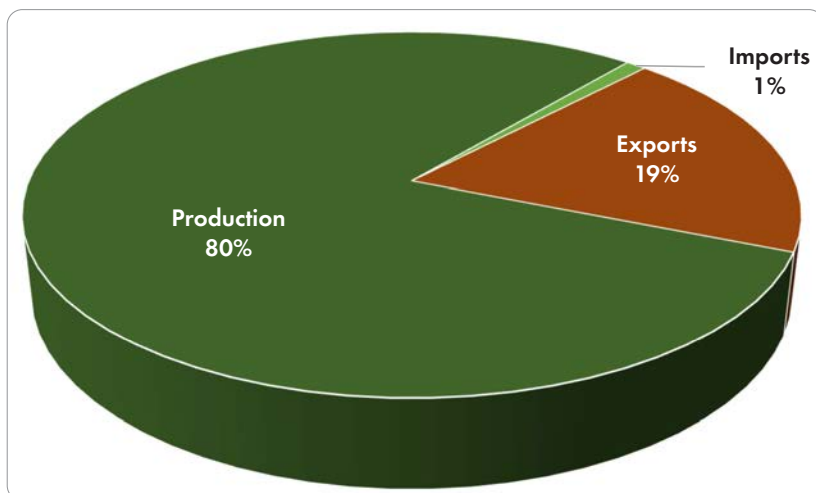
Source: Eskom Integrated Results, 2021

Coal-fired power stations made up 84% of the nominal capacity of the total electricity generated in 2021 (Figure 8 above). The rest of the electricity generated came from a combined contribution by other power stations which accounted for 16% of total nominal capacity with nuclear accounting for 4%, Pumped storage at 6%, Open-cycle gas turbines at 5% and Hydro at 1%. (Figure 8 above).

1.4 Coal

The primary indigenous energy resource base in South Africa is coal. Coal is the primary fuel most used worldwide, making up over 36% of the fuel used in the production of electricity worldwide. Coal provides about 77% of South Africa's energy needs (DMRE, 2021). South Africa exported 19% of coal while it imported 1% in 2021. Domestic production contributed 80% to the total coal primary supply as per Figure 9 below.

Figure 9: Coal primary supply, 2021



Source: DMRE Energy Balances, 2021

1.5 Renewables

Renewable Energy Independent Power Producer Procurement Programme (REIPPPP)

The goal of South Africa's Renewable Energy Independent Power Producer Procurement Program (REIPPPP) is to promote the growth of renewable energy projects. The South African government introduced it in 2011 as part of its goal to lessen reliance on fossil fuels and increase the amount of renewable energy in the nation's energy mix. Remember that the application might have had upgrades or modifications since then. The REIPPPP is also aimed at bringing additional megawatts into the country's electricity system through private sector investment in onshore wind, photovoltaic, concentrated solar power (CSP), biomass, landfill gas and small hydro technologies.

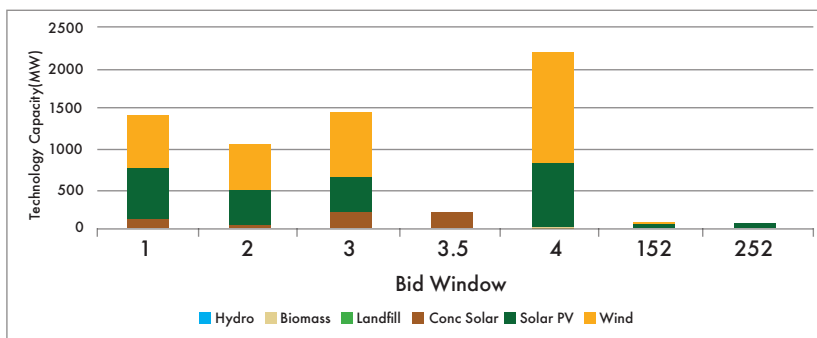
The following are the principal functions and goals of the REIPPPP:

- **Encouraging the Development of Renewable Energy Projects:** The main objective of REIPPPP is to encourage the growth of renewable energy projects, such as small-scale hydroelectric, wind, solar, and biomass projects. South Africa hopes to accomplish these goals by diversifying its energy sources and lowering greenhouse gas emissions.
- **Private Sector Involvement:** Independent power producers (IPPs) are invited to submit proposals for renewable energy projects as part of the program's efforts to promote private sector involvement. This makes it possible for private businesses to finance and run renewable energy projects.
- **Competitive Bidding procedure:** REIPPPP uses a competitive bidding procedure in which prospective developers submit proposals. The bidders who offer the highest value in terms of cost, technical viability, and local economic development are chosen to carry out the projects.
- **The program has a strong emphasis on the creation of jobs and economic development, especially in local communities.** Implementing socio-economic development projects including community infrastructure projects and skill development programs is often a requirement for successful bidders.
- **System Integration:** In order to provide a steady and dependable supply of electricity, REIPPPP seeks to integrate renewable energy projects into the country's electrical system. Coordination with the national utility and pertinent authorities is required to handle the technical and legal facets of grid integration.

- **Reaching Renewable Energy Goals:** By enabling the deployment of a predetermined quantity of renewable energy capacity over predetermined time periods, the program helps South Africa achieve its larger renewable energy targets as stated in the Integrated Resource Plan (IRP).

The REIPPP program constitutes one of the energy mixes as outlined in the National Development Plan and the Integrated Resource Plan 2010. The energy supply capacity per technology in 2021 was dominated by onshore wind power at 52%, followed by photovoltaic power at 36% and CSP (9%), respectively.

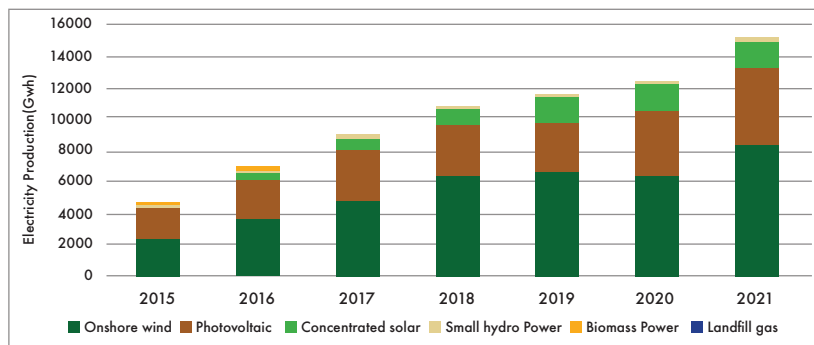
Figure 10: Programme capacity in windows 1, 2, 3 and 4



Source: Department of Mineral Resources and Energy (DMRE), 2021

In 2021, the REIPPPP power production was dominated by onshore wind power at 55%, followed by photovoltaic power at 33%. Concentrated solar power contributed 11% while small hydropower contributed 1% to electricity produced through the REIPPPP. A total of 15 135 GWh was produced in 2021.

Figure 11: Electricity production through REIPPPP, 2015 – 2021



Source: Department of Mineral Resources and Energy (DMRE), 2021

1.6 Nuclear

Globally, many countries have different approaches to nuclear energy. While some significantly rely on it to supply their electrical demands, others have phased it out or are more cautious. Because nuclear power produces electricity steadily and continuously, it can be used to generate base-load power and improve energy security by lowering reliance on fossil fuels. But issues like high upfront capital expenditures worries about nuclear accidents and safety, and handling radioactive waste have resulted in different public attitudes and different government actions.

Nuclear energy in 2021 contributed about 4% to the total energy supply. Koeberg is the only nuclear power station in Africa and has the largest turbine generators in the Southern Hemisphere and is the most southerly-situated nuclear power station in the world. Koeberg ranks amongst the safest of the world's top-ranking PWRs of its vintage and is the most reliable Eskom power station. The station is located 30 km north of Cape Town (Eskom, 2022).



2. ENERGY DEMAND

Energy is the foundation of the South African economy, propelling industrialization, fostering economic expansion, and raising general productivity. The energy landscape of the nation is varied and includes nuclear power, renewable resources, and coal, to a lesser degree. South Africa has historically relied largely on coal to generate electricity, but in recent years, diversification and the integration of renewable energy sources have gained importance. Mining, industrial, and service industries all depend on a consistent and reasonably priced energy source. Sufficient energy infrastructure promotes economic competitiveness and innovation in addition to ensuring that industries run smoothly.

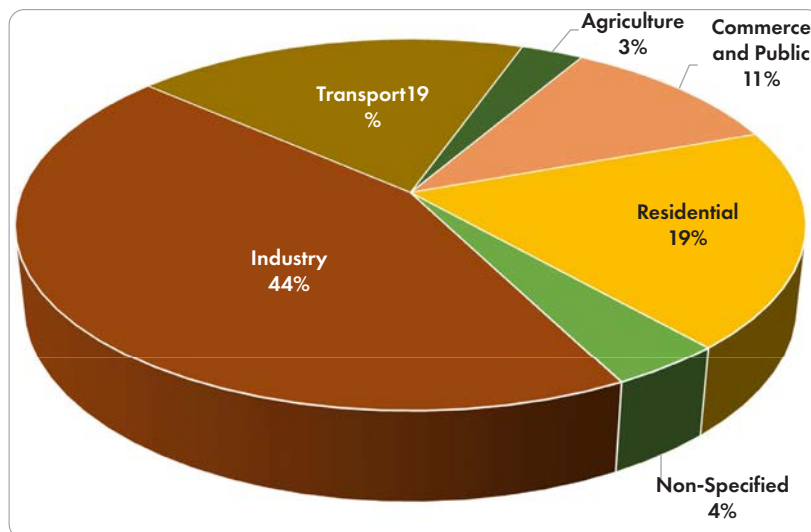
Furthermore, in line with worldwide trends in the use of renewable energy, there is a greater emphasis on shifting to greener and more sustainable energy solutions as South Africa seeks to address environmental issues and lower its carbon footprint.

Obstacles like sporadic power supplies, outdated infrastructure, and the requirement for large investments highlight how crucial policy formulation and execution are to strengthen the energy industry. The dedication to diversifying the energy mix and promoting a more robust and sustainable economy is demonstrated by programs like the Renewable Energy Independent Power Producer Procurement Program (REIPPPP). The significance of energy as the foundation of the South African economy remains crucial as the nation navigates these difficulties and moves toward a more balanced and environmentally friendly energy portfolio.

The South African economy's reliance on energy continues to be essential for attaining long-term stability, competitiveness, and resilience in the face of shifting global energy dynamics.

The share percentage of energy consumed by various sectors in the economy is depicted in Figure 12 below. The five sectors identified in this report are industrial, transport, agriculture, residential, commerce and public services. The sector “non-specified” refers to unaccounted energy (energy that has not been classified into a specific sector).

Figure 12: Energy demand by sector, 2021

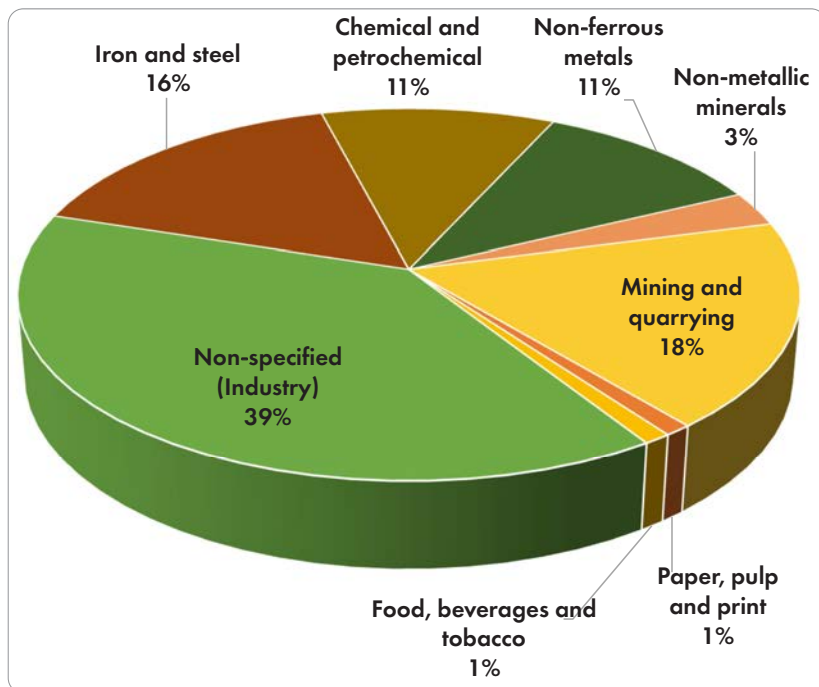


Source: DMRE Energy Balances, 2021

2.1 Industrial sector

Energy demand in the industrial sector has been dominated by the mining and quarrying sub-sector (18%) and the Iron and steel at 16%. Petrochemical and non-ferrous metals contributed 11% each respectively. The remaining sub-sectors contributed 3% or less to the total energy consumption in the Industrial sub-sectors for the year 2021. (Figure 13 below).

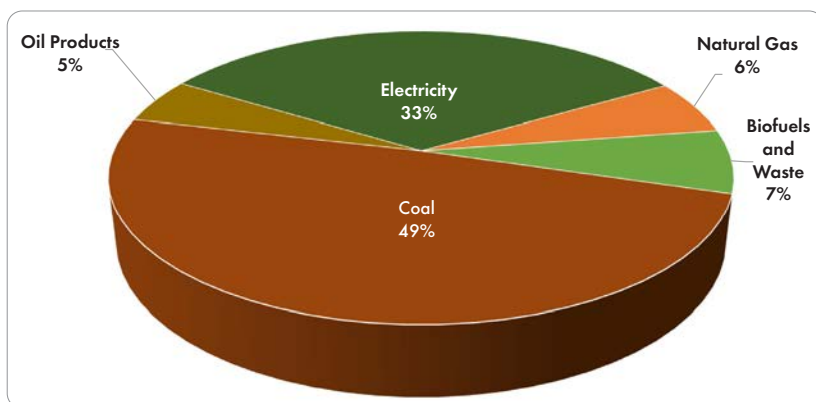
Figure 13: Energy demand by industrial sub-sectors, 2021



Source: DMRE Energy Balances, 2021

Figure 14 below shows that coal (49%) and Electricity (33%) have been the most consumed sources of energy in the industrial sector for the year 2021. This was followed by Biofuels and waste at 7% and Natural Gas at 6%. Oil products accounted for 5% to the total energy consumption in the Industrial sector for the year 2021.

Figure 14: Energy demand in the industrial sector, 2021

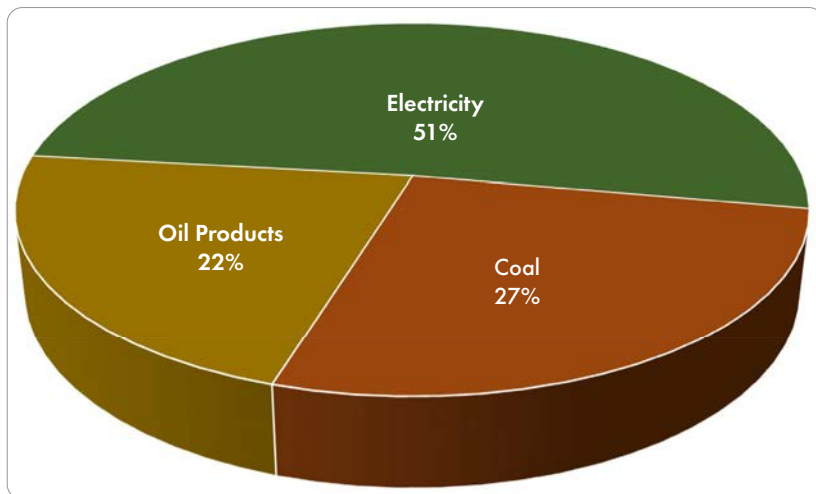


Source: DMRE Energy Balances, 2021

2.1.1 Mining

The mining sector is one of the major consumers of energy in South Africa, particularly electricity. The mining industry is a significant energy consumer in South Africa and is vital to the country's economy. Owing to the abundance of mineral resources in the nation, which includes coal, diamonds, gold, and platinum, the mining sector is highly dependent on energy for a number of operations, including extraction, processing, and transportation. Deep-level mining and mineral beneficiation are two examples of energy-intensive procedures that add significantly to the sector's energy requirements. The mining industry and South Africa's energy landscape are closely related, as seen by the reliance on energy-intensive operations. Finding effective and cleaner energy solutions for the mining industry is crucial as the country works to address its energy problems and move toward more sustainable energy sources in order to maintain both environmental responsibility and economic prosperity. Electricity contributed 51% to the total energy consumed in the mining sector in 2021, while Oil products and coal contributed 22% and 27% respectively in 2021. The mining sector consumed approximately 198,949 TJ of energy in total in the year 2021. (Figure 15 below).

Figure 15: Energy demand in mining, 2021

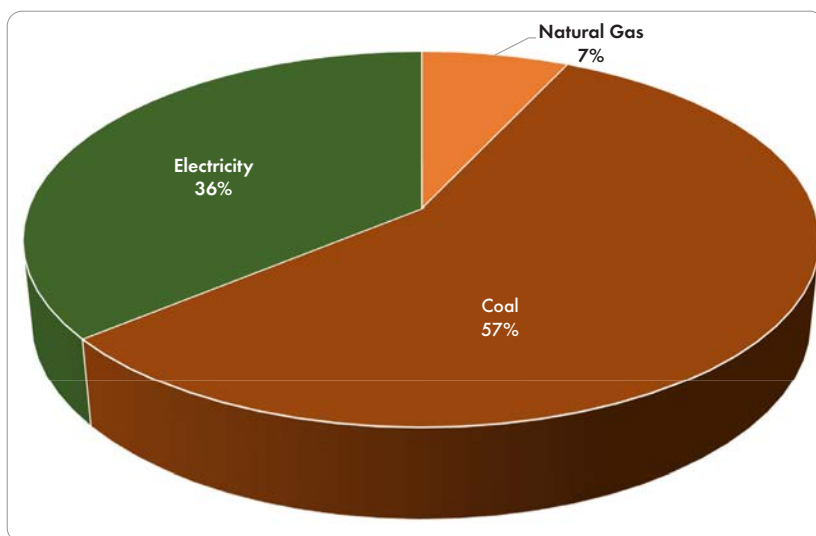


Source: DMRE Energy Balances, 2021

2.1.2 Iron and steel

In many economies, including South Africa, the iron and steel sector is a notable power consumer. This industry is vital to the nation's industrial landscape since it supports manufacturing, building, and the development of infrastructure. Smelting ore, refining, and running large machinery are just a few of the activities in the iron and steel industry that depend on electricity as a necessary input. The energy-intensive nature of these operations highlights the iron and steel industry's significant electricity requirements. The main energy carriers in the Iron and steel industry comprise of electricity, coal and Natural gas. Coal contributed the most at 57% to the final energy demand in the Iron and Steel industry. Electricity followed with a contribution of 36% while Natural gas contributed 7%. The iron and steel industry utilised a total energy of 176 768 TJ in 2021. (Figure 16 below).

Figure 16: Energy demand in Iron and Steel industry, 2021



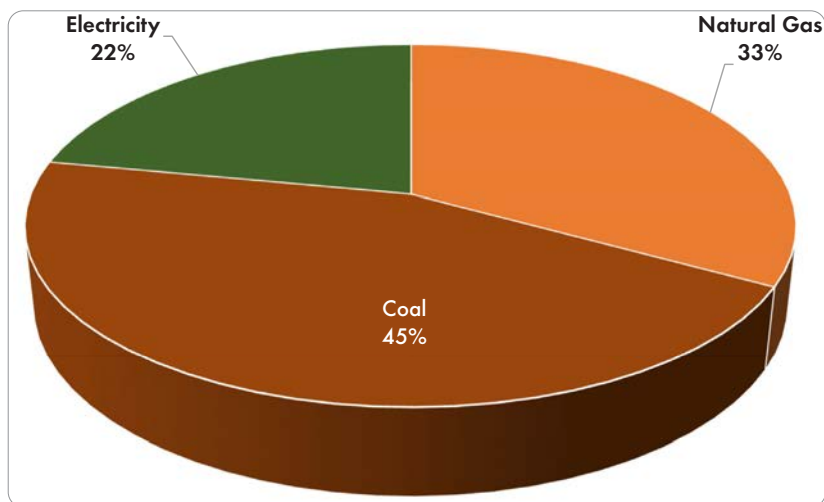
Source: DMRE Energy Balances, 2021

2.1.3 Chemicals

South Africa's chemical sector plays a major role in the industrial and economic growth of the nation by consuming a large amount of electricity. This industry includes a broad range of operations that all require significant energy inputs, such as fertilizers, polymers, and petrochemicals. Chemical reactions including synthesis, separation, and refining as well as the operation of several pieces of machinery and equipment depend on electricity. The fact that the chemical industry depends on power highlights how important it is to South Africa's manufacturing sector.

The chemical industry in South Africa intensively uses gas and coal in the manufacturing process. Coal contributed 45% while electricity and Natural gas contributed a share of 22% and 33% respectively. A total energy of 128 736 TJ was consumed in the chemical and petrochemical industry in the year 2021. Figure 17 depicts the energy demand in the chemical and petrochemical industry.

Figure 17: Energy demand in Chemical and Petrochemical industry, 2021



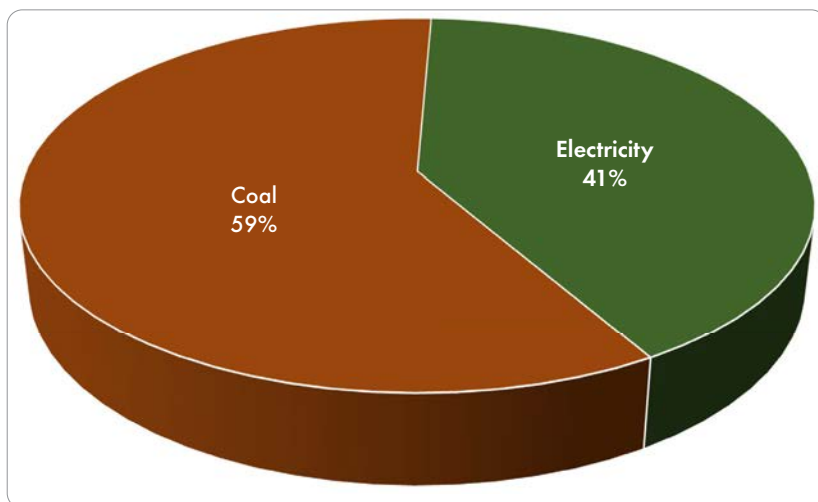
Source: DMRE Energy Balances, 2021

2.1.4 Non-Ferrous Metals

An important user of energy in South Africa is the non-ferrous metals sector, which produces metals like copper, aluminum, and nickel. These metals are essential components used in many manufacturing industries, such as electronics, automotive, and construction. Electricity is required in the non-ferrous metals sector for operations including smelting, refining, and forming raw materials into finished goods. These procedures require a lot of energy, which raises the sector's need for power significantly.

The total energy consumption in the non-ferrous metal industry in 2021 amounted to 127,238 TJ. Coal was consumed most in the non-ferrous metals and amounted to 74 965 TJ, which contributed 59% and electricity contributed 41% to the total energy demanded consuming an overall total of 52 018 TJ in 2021 as per Figure 18 below.

Figure 18: Energy demand in non-ferrous metals, 2021



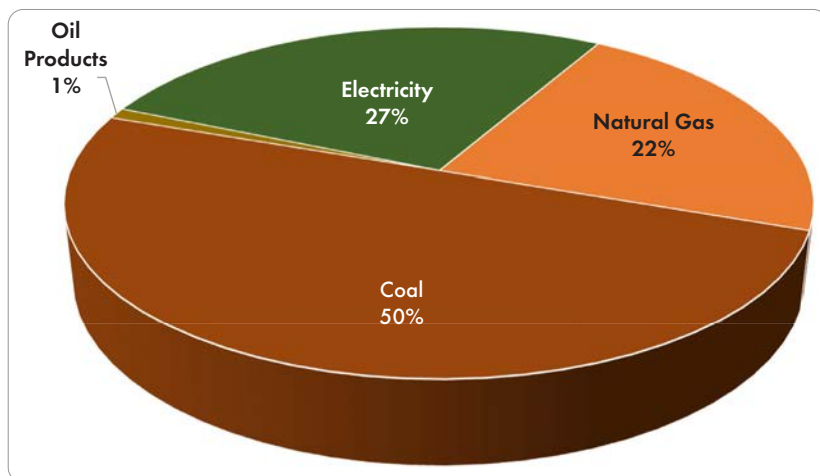
Source: DMRE Energy Balances, 2021

2.1.5 Non-Metallic Minerals

Electricity is heavily consumed by South Africa's non-metallic minerals industry, which includes operations like quarrying, processing, and producing goods like cement, glass, and ceramics. This industry is essential to the development of infrastructure, building, and a number of manufacturing processes. Electricity is essential for running crushers, mills, kilns, and other extraction and processing machinery in the production of non-metallic minerals. Because of this, the sector uses a lot of electricity, which is indicative of the vital function that electricity plays in converting raw materials into final commodities.

In 2021, coal dominated the non-metallic minerals sector with 50%, followed by electricity contributing 27%, Natural gas (22%), and 1% for oil products. The total energy consumed by the non-metallic minerals sector was 29,000 TJ in 2021. (Figure 19 below).

Figure 19: Energy demand in non-metallic minerals, 2021



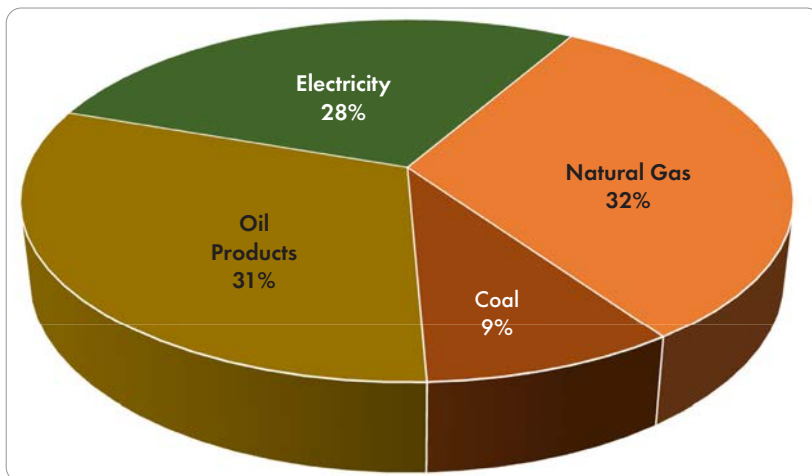
Source: DMRE Energy Balances, 2021

2.1.6 Food, Beverages and Tobacco

Considering the wide range of activities, it engages in, such as food processing, beverage production, and tobacco manufacturing, the food, beverage, and tobacco business in South Africa is a significant power user. In every step of these operations, from cooking and refrigeration to packaging and delivery, electricity is a necessary component. Electricity is essential to the sector to run machinery, keep quality control, and guarantee that strict hygienic regulations are followed. The sector's electricity usage signifies its importance in supplying necessities to both local and foreign markets, making it a crucial contributor to the nation's economy.

The total energy consumed by the food, beverages, and tobacco sector was 9296 TJ in 2021 with Natural gas and Oil products contributing 32% and 31% respectively. Electricity contributed 28% while coal contributed 9% to the total energy consumed in the sector as depicted in Figure 20 below.

Figure 20: Energy demand in food, beverages and tobacco, 2021



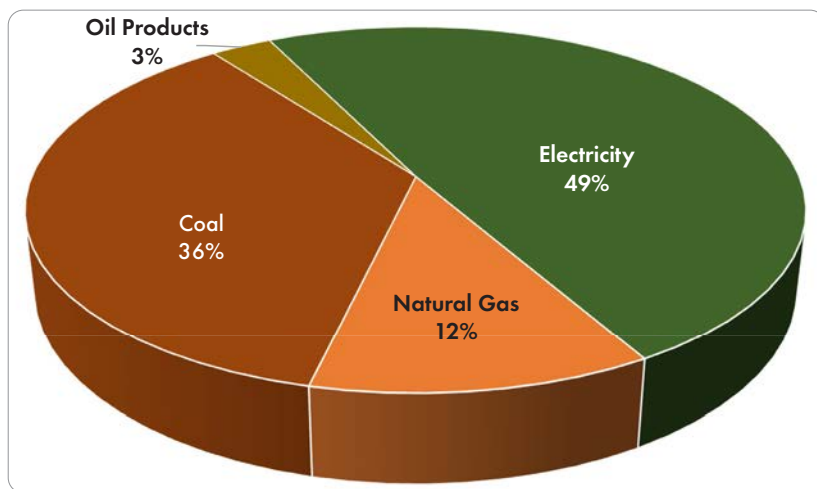
Source: DMRE Energy Balances, 2021

2.1.7 Pulp and Paper

The South African pulp and paper industry, which is essential to the production of paper and related goods, uses a lot of electricity. From the pulping of wood and other raw materials to the several stages of paper manufacturing, such as drying, printing, and refining, electricity is necessary for the entire production process. The necessity to power machinery, heating systems, and other necessary equipment has resulted in the industry's strong reliance on electricity. The industry's electricity usage is indicative of its significance in bolstering other sectors like publishing, packaging, and education, given its contributions to writing materials, printing, and packaging.

The paper, pulp and print industry mostly used electricity as their main source of energy in 2021. Electricity contributed 49% and coal contributed 36%. Natural gas and oil products contributed the lowest at 12% and 3% respectively. This industry consumed 7587 TJ in total as per Figure 21 below.

Figure 21: Energy demand in paper, pulp and print industry, 2021.



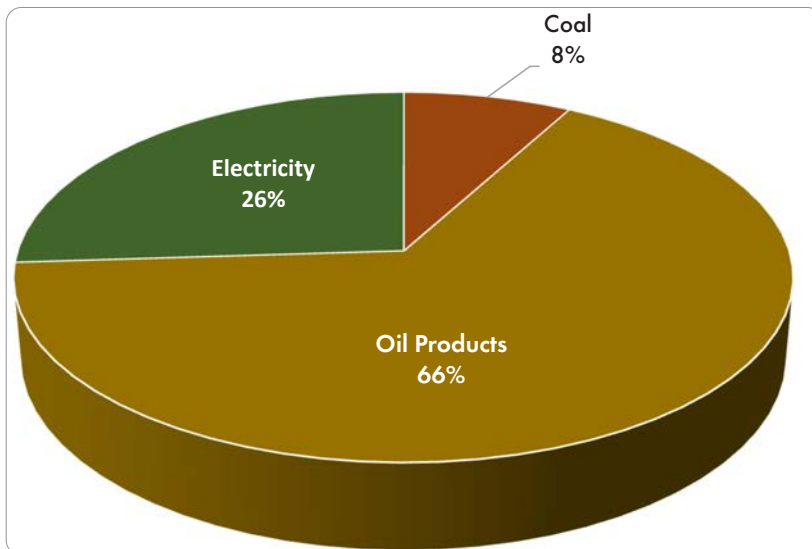
Source: DMRE Energy Balances, 2021

2.2 Commerce and Public Services Sector

In South Africa, the large consumer of electricity is the Commerce and Public Services sector, which includes a broad range of activities such as offices, retail, government services, and public institutions. Powering offices, lighting, HVAC systems, electronic devices, and other vital infrastructure is a critical function of electricity in this industry. Electricity consumption in this sector is important for maintaining the smooth running of enterprises, government functions, and public facilities since it is the foundation of the economy and public services. In order to maximize efficiency and reduce environmental effects, the task is to optimize energy use and implement sustainable methods.

The commercial sector comprises of financial services, information technology, retail, tourism, and services industry. The public services sector comprises of government and quasi-government institutions which provide goods and services to the public, mainly for free. Electricity accounted for 26% of the total energy consumed while Oil products were the highest and contributed 66% to the total energy consumed. Coal was the lowest energy consumed in the sector and accounted for only 8%. The total energy consumed in by this sector is approximated to 288 621 TJ in 2021 as per Figure 22 below.

Figure 22: Energy demand in the commerce and public services sector, 2021



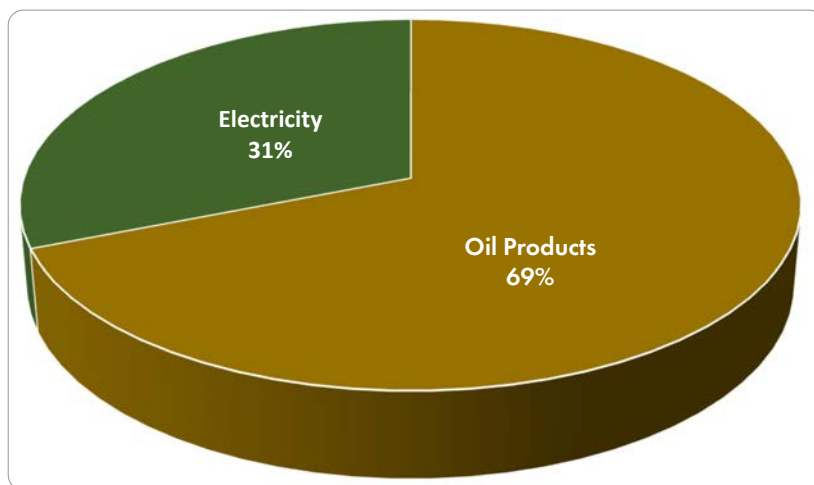
Source: DMRE Energy Balances, 2021

2.3 Agricultural and Forestry Sector

Crop cultivation, animal husbandry, forestry, and other related industries are all included in South Africa's vast and important agricultural and forestry sector. Even while this industry may not use electricity as much as other industrial sectors, it is nevertheless quite important to many different operations. Irrigation systems, processing facilities, storage, and equipment operations all require electricity. Technology-driven methods in agriculture, like automated systems and precision farming, increase the need for electricity. In the forestry industry, electricity is necessary for sawmill management and timber processing. For the Agricultural and Forestry sector to remain resilient and a driving force in South Africa's economy while minimizing its negative effects on the environment, it is imperative that sustainable and energy-efficient practices be used in conjunction with the electricity needed to increase productivity and efficiency.

As shown in figure 23 below, the agriculture and forestry sector used oil products as the main energy carrier which contributed 69% to the total energy consumed for 2021. Oil products alone amounted to a consumption of 48081 TJ. This is a result of the transportation of agricultural raw materials, feeds, intermediaries, and finished products from farms to various market areas. Electricity accounted for 31% to the total energy consumed in the agricultural and Forestry sector and amounted to 21 649 TJ in 2021. (Figure 23 below).

Figure 23: Energy demand in the agriculture and forestry sector, 2021



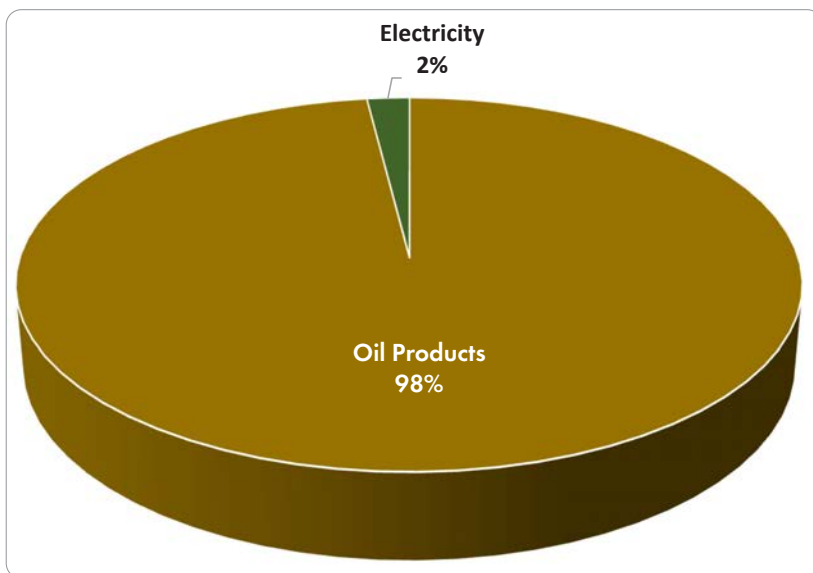
Source: DMRE Energy Balances, 2021

2.4 Transport Sector

In South Africa, the transportation sector is a significant energy user, mostly using liquid fuels like gasoline and diesel. The transportation industry is changing due to the growing popularity of electric cars (EVs) and improvements in electric public transportation, even though the sector's direct power consumption may not be as high as in certain other industries. The increasing popularity of electric vehicles such as cars, buses, and trains is leading to an increase in the need for power. The effective integration of electricity into the transportation sector is contingent upon the development of charging infrastructure, the design of electric vehicles that are efficient, and the implementation of laws that are favourable.

The transport sector, which uses most of oil products, consumed 478 636 TJ in 2021. The majority of this was used on the road at 88%. Oil products accounted for 98% while electricity accounted for 2% of the energy demanded in this sector for the year 2021. A total of 486 836 TJ was consumed in 2021. (Figure 24 below).

Figure 24: Energy demand in the transport sector, 2021



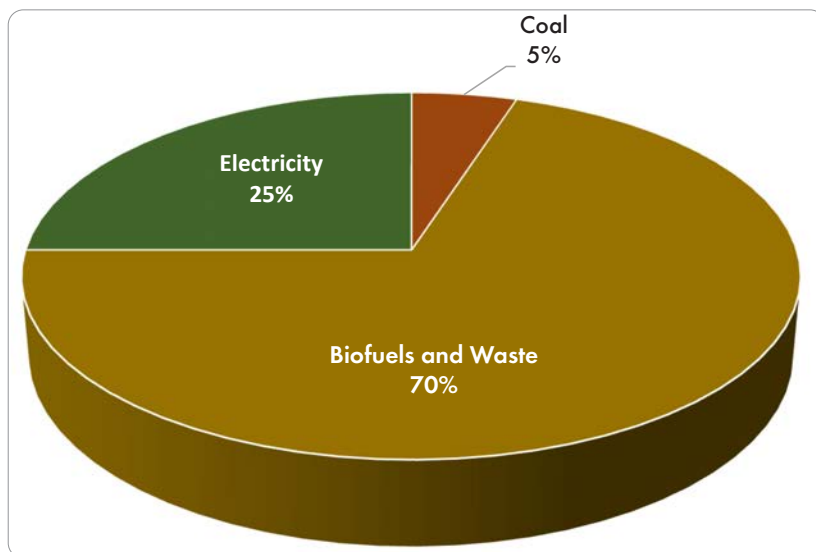
Source: DMRE Energy Balances, 2021

2.5 Residential Sector

South Africa's residential sector, which reflects the various energy needs of homes, accounts for a sizeable amount of the country's electricity consumption. In homes, electricity is necessary to run appliances, lights, air conditioning, heating, and other electronics. The country's residential sector is becoming more and more dependent on power as a result of growing urbanization and electrification. To lower the overall electricity demand, the challenge is to promote energy efficiency, encourage the use of energy-efficient equipment, and adopt smart home technology. Furthermore, programs such as home solar power systems help create a domestic energy landscape that is more sustainable.

The residential sector accounted for 19% of the total energy consumption in 2021. The consumption of electricity amounted to 25% of the total energy consumed in the sector. Biofuels and waste accounted for 70% while coal accounted for 5%. The total energy consumed in the sector amounted to 367 968TJ of energy in total. (Figure 25 below).

Figure 25: Energy demand in the residential sector, 2021



Source: DMRE Energy Balances, 2021

NB: Due to misaligned mappings of the raw data on consumption to the ISIC codes, the Department of Mineral Resources and Energy estimated the residential consumption of coal as one third of sales to merchants and domestic markets. This estimate kept on increasing over the years, which is contrary to the increasing number of residential electricity connections.

It was subsequently found necessary to estimate the actual consumption in this sector using methodologies that take into account the actual residential coal end-use data (from the General Household Survey:2023) carried out by Statistics South Africa (StatsSA). The methodology was based on the average (in Rands) spent each month by sampled households. This average was used to calculate tons of coal consumed by dividing the total expenditure by the average coal price. Inferring these to the total population resulted in a massive drop in coal consumption in the residential sector.

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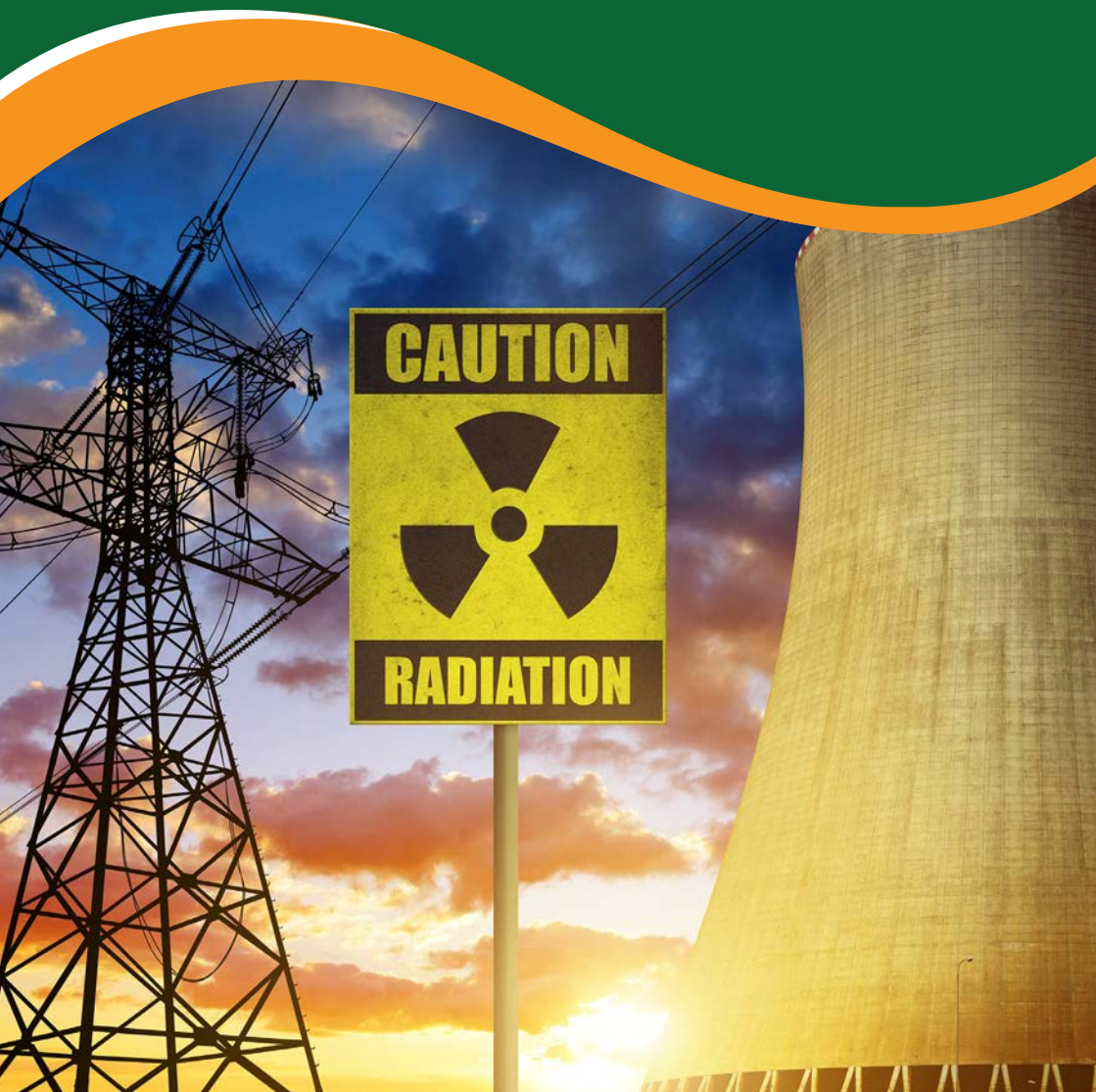
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